

# Switch Selection Strategy Report: Transitioning to Solid-State Pulse Discharge Systems

## 1. Strategic Objectives: Repeatability and ESL Extraction Accuracy

The primary mission of this transition is to replace legacy mercury-wetted relays and aging thyatron technologies with modern semiconductor alternatives. From a strategic sourcing perspective, the reliance on thyratrons presents an escalating risk as these components become increasingly difficult to source and maintain. In our industrial film capacitor testing environment, the pivot to solid-state systems is driven by the need for superior **repeatability (Gauge R&R)** and environmental compliance.

A secondary but vital architectural objective is the **precise extraction of Equivalent Series Inductance (ESL)**. To achieve this, our system must minimize parasitic inductance within the discharge path, as standard mechanical housings often introduce inductive noise that masks the capacitor's true characteristics. This report prioritizes "low inductive housing" designs and "integrated driver units" to shorten current paths. By adopting semiconductor switches with monolithic integration, we can significantly reduce the loop area and ensure the integrity of high-frequency measurements.

## 2. Comparative Technical Analysis of Switching Technologies

The following table summarizes the semiconductor alternatives analyzed for pulsed discharge applications. Data is synthesized from industrial benchmarks and recent research into overvoltage triggering modes.

| Technology Type     | Forward Blocking Voltage      | di/dt Capability            | Primary Application Logic                                                                  |
|---------------------|-------------------------------|-----------------------------|--------------------------------------------------------------------------------------------|
| Thyristors (SCR)    | Up to 8500 V                  | Low (< 1.5 kA/ $\mu$ s)     | High current (120 kA), long pulse discharge, and dump switches.                            |
| GTO-Like Thyristors | Up to 4500 V                  | High (Up to 50 kA/ $\mu$ s) | High-speed magnetic forming; fast (~1 $\mu$ s) switch-on using interdigitated segments.    |
| IGBT Modules        | Up to 6500 V                  | Moderate (~5 kA/ $\mu$ s)   | Applications requiring high switch-off capability and pulse currents of ~1 kA.             |
| SiC MOSFETs         | 1.7 kV - 11 kV+ (Marx Driver) | Ultra-High (>10 kV/ns)      | High-voltage nanosecond pulses; optimized for gate-boosting and impact-ionization drivers. |

|                              |             |                    |                                                                                               |
|------------------------------|-------------|--------------------|-----------------------------------------------------------------------------------------------|
| <b>GaN (Gallium Nitride)</b> | Moderate    | Exceptionally High | <b>Premier choice</b> for ultra-low inductance and high-precision ESL extraction.             |
| <b>SAS / PCSS</b>            | Specialized | Sub-nanosecond     | Specialized high-speed alternatives for sub-nanosecond research and specialized physics work. |

**3. Primary Recommendation: GaN for Low-Inductance Performance**

For applications focusing on high-precision ESL extraction, **Gallium Nitride (GaN)** is the primary recommendation. GaN technology offers the compact footprint and switching speeds necessary to minimize the parasitic inductance that traditionally plagues film capacitor testing.

**Layout Principles and Inductance Minimization**

To maximize GaN’s performance, the architecture must minimize the inductive path between the driver and the device. We recommend the following layout principles derived from advanced power electronics packaging:

- **Monolithic Integration:** Where possible, utilize "Reverse Conducting" designs—similar to the monolithic integration of freewheeling diodes in thyristors—to minimize the inductive path between the switch and the diode.
- **Driver Proximity:** The gate driver must be placed in immediate proximity to the GaN switch on the same PCB plane. This minimizes the driver loop area and prevents signal degradation.
- **Minimized Power Loop:** Using small-package footprints (such as D2PAK) allows for a reduction in the total current loop area, which is the primary driver of parasitic inductance. Reducing this area is critical to preventing the "masking" of the capacitor's ESL during high-frequency discharge.

**4. High-Voltage Strategy: Gate-Boosted Marx Architectures**

When discharge requirements exceed the voltage limits of single-stage GaN components, we propose a multi-stage architecture utilizing a Marx generator.

**Gate-Boosting Driver Methodology**

High-voltage precision is achieved through a custom **Gate-Boosting** driver stage. This stage utilizes **Silicon Carbide (SiC) MOSFETs** triggered by fast gate drivers. These SiC devices are operated in a gate-boosted mode to provide a fast slew rate (>1 kV/ns) and

substantial overvoltage. This driver stage provides the necessary energy to trigger the subsequent Marx generator stages with sub-nanosecond precision.

### Multi-Stage Marx Generator with D2PAK Thyristors

Contrary to standard MOSFET-based Marx designs, our high-voltage output stage is a 4-stage Marx generator composed of **small, 1.2 kV rated D2PAK thyristors** (specifically the VS-16TTS12-M3). This configuration effectively charges capacitors in parallel and discharges them in series. By utilizing these COTS thyristors in impact-ionization mode, the generator can achieve output pulses exceeding 11 kV with a  $dU/dt$  of 13.3 kV/ns.

### Impedance Grading and FEMM Analysis

To maintain current continuity and prevent signal reflections that could stress the thyristors, we implement **Impedance Grading**:

- **Design Approximations:** Trace widths are calculated using Finite Element Method Magnetics (FEMM) models to match transmission line impedances across the PCB.
- **Transition Management:** The design follows a transition from 3.5  $\Omega$  at the initial SiC driver stage to 24.5  $\Omega$  at the final Marx stage. While an initial impedance mismatch is intentionally introduced at the first stage to facilitate triggering via positive reflection, subsequent stages must be matched to minimize high-voltage spikes and maintain pulse integrity.

## 5. Engineering for Repeatability: Impact-Ionization and Reliability

The transition to solid-state systems significantly improves Gauge R&R through the utilization of **Impact-Ionization Mode**.

- **Trigger Mechanism:** By applying an overvoltage at least twice the static breakdown and a slew rate exceeding 1 kV/ns, we create an ionization front within the semiconductor. This allows for sub-nanosecond triggering that is far more predictable than the stochastic behavior of mechanical relays or gas-filled spark gaps.
- **Environmental Reliability:** Unlike mercury-wetted relays, these semiconductor switches are mercury-free, offer flexible mounting positions, and require zero maintenance.
- **Operational Lifetime:** Solid-state switches provide a much longer lifetime (exceeding 20,000 shots in high-stress conditions) compared to the limited cycle counts of mechanical alternatives, ensuring the test system remains stable and repeatable over years of operation.

6. Implementation Matrix and Conclusion

The following framework provides the final decision-making guide for selecting specific switches:

| Application Need                | Primary Technology    | Configuration Strategy                                                        |
|---------------------------------|-----------------------|-------------------------------------------------------------------------------|
| Low Inductance / Precision ESL  | GaN                   | Compact footprint, integrated driver, monolithic diode pathing.               |
| High Voltage (>10 kV) Discharge | SiC + D2PAK Thyristor | SiC-based Gate-Boosting driver with a multi-stage Marx Generator.             |
| High Current / Magnetic Forming | GTO-Like Thyristor    | Integrated Gate Controlled Thyristor (IGCT) or Reverse Conducting Thyristors. |

Conclusion

To ensure a successful transition, we must prioritize **"Ground-Truth" calibration** using **capacitive probes**. These non-contact sensors preserve signal integrity and protect measurement equipment from high-voltage transients. Architects must account for the **frequency-dependent limitations of FR4 PCB material**, which can influence capacitive probe accuracy at high frequencies. When designed with these constraints in mind, the new solid-state pulse discharge system will provide the robust, repeatable data necessary for modern capacitor manufacturing and testing.